INFOTACT SOLUTIONS PROJECT DOCUMENT – ADVANCED DATA ANALYTICS (VOL. II)

Real-World Applications - Hands-on System Building

To continue expanding your advanced Data Analytics portfolio, this volume moves entirely away from standard relational dashboards (like sales charts) and focuses on the most highly sought-after, specialized analytics niches: **Streaming Graph Analytics**, **Geospatial Mobility**, and **Process Mining**.

Here are three more highly advanced, deeply technical Data Analytics projects for a senior-level portfolio.

Project 1 - "FinGraph": Real-Time Fraud Syndicate Analytics

Domain: FinTech & Anti-Money Laundering (AML)

Problem Statement: *Standard fraud analytics rely on relational tables and isolated rules (e.g., "flag transactions over $10,000"). They completely miss "smurfing" or complex money laundering syndicates, where 100 seemingly unrelated accounts send $9,900 each through a maze of intermediary accounts to evade detection.*

Use Case: A risk analyst at a major bank views the **FinGraph** dashboard. Instead of reading a list of suspicious transactions, they see a real-time, self-updating network graph. Streaming transaction data flows through Kafka and is processed by Apache Flink into Neo4j. The dashboard instantly flags a "Starburst" pattern: 50 distinct accounts, which share no common IP addresses, are all funneling micro-transactions into a single offshore shell company. The analyst freezes the entire syndicate with one click.

Key Modules:

- **Stream Processor (Apache Flink & Kafka):** Ingests live mock transaction data and performs real-time stateful aggregations before pushing to the graph.
- **Graph Database (Neo4j):** Stores transactions not as rows, but as interconnected nodes (Users, Accounts) and edges (Transactions, Shared IPs).
- **Graph Algorithms (Neo4j GDS):** Uses native Graph Data Science algorithms like PageRank and Weakly Connected Components (WCC) to score the "centrality" of suspicious accounts.
- **Network Visualization (React & Linkurious / Superset):** A highly interactive, node-based dashboard allowing analysts to visually untangle the web of illicit money flows.

Week-wise Development Plan:

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Week	Data Engineering (Kafka, Flink, Neo4j)	Analytics & UI (Cypher, React/Superset)
Week 1	**Ingestion Setup:** Build a Python simulator generating realistic, nested syndicate transaction networks. Stream this data into Kafka topics.	**Graph Schema:** Design the Neo4j schema (Nodes: Person, Account, Bank. Edges: TRANSFERRED_TO).
Week 2	**Stream Processing:** Write Apache Flink jobs to consume the Kafka stream, clean the data, and continuously upsert the nodes/edges into Neo4j in real-time.	**Cypher Queries:** Write complex Cypher queries to detect circular money flows (A → B → C → A) and calculate node risk scores.
Mid-Project Review	**Pipeline Audit:** Prove that a transaction sent via the Python simulator appears as a connected edge in the Neo4j database in under 1 second.	**Query Optimization:** Ensure complex multi-hop Cypher queries execute in under 100ms.
Week 3	**Graph Data Science:** Implement Neo4j GDS algorithms. Run Louvain community detection to automatically group suspicious clusters of accounts together.	**Dashboard UI:** Connect a visualization tool (like Linkurious or a custom React/D3 app) directly to the Neo4j endpoint.
Week 4	**Automation:** Implement a rules engine that triggers automated Slack/Email alerts when a user's Graph Risk Score exceeds a specific threshold.	**Refine & Polish:** Finalize the visual layout, allowing the analyst to expand/collapse nodes and visually trace the money trail.
Final Review	A massive paradigm shift from relational data to complex streaming graph architectures.	An enterprise-grade, visual fraud investigation tool.

Project 2 - "GeoPulse": Hyper-Local Retail Mobility Analytics

Domain: Retail Strategy & Geospatial Analytics

Problem Statement: *Retailers often guess where to open new stores based on static census data. They lack dynamic insights into how foot traffic actually flows through a city throughout the day, often accidentally opening a new store that cannibalizes the sales of their existing locations.*

Use Case: A real estate director for a coffee chain uses **GeoPulse** to evaluate a new store location. The pipeline ingests massive datasets of anonymized mobile GPS pings. Using spatial databases, the system

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overlays these pings onto commercial zoning polygons. The Kepler.gl dashboard generates a dynamic, animated footfall heatmap. The director scrubs through the timeline and realizes that placing "Store B" on 5th Street will intercept and cannibalize 30% of the morning commuter traffic currently walking to their existing "Store A," prompting them to choose a different neighborhood.

Key Modules:

- **Geospatial Data Lakehouse (Snowflake):** Stores massive arrays of X/Y coordinate pings and utilizes Snowflake's native GEOGRAPHY data types.
- **Spatial Transformation (Apache Sedona / PySpark):** Distributed computing framework to execute heavy spatial joins (e.g., checking if 10 million GPS points intersect with 5,000 store polygons).
- **Data Modeling (dbt):** Cleanses the geospatial joins and aggregates the data into hourly "Footfall Traffic" metrics.
- **Mobility Dashboard (Kepler.gl / React):** Uber's open-source geospatial visualization tool, embedded into a React app for rendering massive 3D hexbins and movement arcs.

Week-wise Development Plan:

Week	Spatial Data Engineering (Snowflake, PySpark)	Geospatial BI (dbt, Kepler.gl)
Week 1	**Data Synthesis:** Generate a massive mock dataset of anonymized mobile device pings (DeviceID, Lat, Long, Timestamp) mimicking city movement.	**Data Loading:** Set up Snowflake. Create tables utilizing GEOGRAPHY column types and load the raw ping data.
Week 2	**Spatial Joins:** Use Apache Sedona (Spatial Spark) to draw 500-meter "catchment area" polygons around target store locations and intersect them with the ping data.	**ELT Modeling:** Configure dbt to run inside Snowflake. Create models that aggregate total unique visitors per store per hour.
Mid-Project Review	**Spatial Audit:** Prove the PySpark job can successfully calculate intersections for millions of points without running out of memory.	**Data Integrity Check:** Ensure dbt compiles successfully and the resulting tables accurately reflect morning vs. evening traffic spikes.
Week 3	**Cannibalization Logic:** Write complex SQL in dbt to track unique DeviceIDs over time—determining what percentage of devices visit multiple store locations.	**Map UI:** Integrate Kepler.gl into a React frontend. Connect it to the Snowflake aggregated tables.

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Week 4	**Automation:** Orchestrate the PySpark and dbt jobs using Apache Airflow to run daily updates.	**Refine & Polish:** Configure the Kepler.gl map to render 3D H3 Hexagons (Uber's grid system) and animate the foot traffic over a 24-hour slider.
Final Review	A highly specialized data pipeline mastering the complexities of geospatial coordinate mapping at scale.	A visually stunning, interactive 3D map driving multi-million dollar real estate decisions.

Project 3 - "CareFlow": Clinical Pathway Process Mining

Domain: Healthcare Operations & Process Analytics

Problem Statement: *Hospital administrators don't know why Emergency Room wait times are so high. Standard BI dashboards show the "average wait time," but they cannot map the chaotic, real-world sequence of events a patient actually experiences, obscuring the true operational bottlenecks.*

Use Case: A hospital administrator uses **CareFlow**. The pipeline ingests raw, timestamped event logs from the hospital's Electronic Health Record (EHR) system (e.g., Patient A: Triage at 10:00 -> X-Ray at 10:15 -> Triage at 10:30). Using Process Mining algorithms, it generates a visual "Spaghetti Diagram" of actual patient flows. The dashboard highlights a massive, hidden bottleneck: 40% of patients are being sent *back* to Triage after an X-Ray due to a specific missing paperwork form. The administrator updates the intake form, instantly reducing average wait times by 20 minutes.

Key Modules:

- **Data Warehouse (Google BigQuery):** Highly performant storage for massive volumes of chronological healthcare event logs.
- **Process Mining Engine (PM4Py / Python):** Python library dedicated to discovering process models (Petri nets, Directed Graphs) directly from raw timestamped event data.
- **Data Transformation (dbt):** Pre-processes messy EHR logs to ensure strict standard formatting (Case_ID, Activity_Name, Timestamp).
- **Process Dashboard (Microsoft PowerBI / Looker):** A BI report utilizing custom visual plugins (like process flowcharts) alongside standard operational KPIs.

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Week-wise Development Plan:

Week	Event Log Engineering (BigQuery, dbt)	Process Mining & BI (PM4Py, PowerBI)
Week 1	**EHR Simulation:** Build a Python script generating mock hospital event logs, deliberately coding in operational inefficiencies and loop-backs.	**Warehouse Setup:** Load the raw event logs into BigQuery. Configure dbt for the initial cleaning phase.
Week 2	**Event Normalization:** Write dbt models to standardize the data into an "Event Log" format required by process mining algorithms.	**Process Discovery:** Use the PM4Py library to ingest the dbt output and algorithmically discover the "As-Is" process map of the hospital.
Mid-Project Review	**Pipeline Audit:** Ensure the simulated data flows seamlessly from the Python generator into clean, chronological BigQuery tables.	**Algorithm Check:** Prove PM4Py successfully generates a Directed Graph showing the major paths patients take through the hospital.
Week 3	**Bottleneck Calculations:** Write SQL/Python logic to calculate the average transition time between every single node (e.g., Average time from Triage to Doctor).	**BI Dashboard:** Connect PowerBI to BigQuery. Import custom process mining visuals (like the Power Automate Process Mining visual).
Week 4	**Conformance Checking:** Implement logic to compare the *actual* patient journey against the hospital's *ideal/mandated* journey, flagging non-compliant paths.	**Refine & Polish:** Polish the dashboard, adding drill-down capabilities so administrators can filter the process map by specific patient demographics or doctors.
Final Review	A niche, highly valuable data engineering skill utilizing event-log architecture rather than standard relational tables.	An enterprise healthcare operations dashboard revealing hidden organizational friction.

Common Features Across All Projects

- **Unconventional Data Structures:** These projects require you to abandon standard row/column mindsets. You will engineer networks (Graphs), coordinate geometry (Spatial Polygons), and chronological sequences (Event Logs).
- **Modern Data Stack (MDS):** Heavy reliance on cloud data warehouses (Snowflake, BigQuery) combined with modular ELT transformations using dbt (Data Build Tool), matching modern enterprise standards.

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- **Domain-Specific Visualization:** Standard bar charts are insufficient. These projects utilize specialized BI layers, including interactive force-directed graphs (Linkurious/Neo4j), 3D Geospatial renderers (Kepler.gl), and Process Flow diagrams.
- **Actionable ROI Metrics:** Every dashboard is designed not just to show data, but to drive a specific business action (Freezing an account, moving a store location, altering a hospital intake form).

Overall Project Summary

Project	Target Business Domain	Technology Focus
"FinGraph": Fraud Syndicate Analytics	**FinTech & AML**	Kafka, Apache Flink, Neo4j, Cypher, Graph Data Science
"GeoPulse": Hyper-Local Mobility	**Retail & Real Estate**	Snowflake (Geo), Apache Sedona (Spark), dbt, Kepler.gl
"CareFlow": Clinical Process Mining	**Healthcare Operations**	Google BigQuery, PM4Py (Process Mining), dbt, PowerBI

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